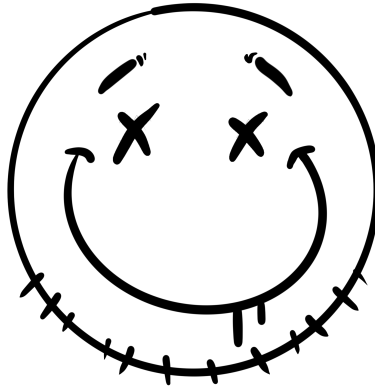




Lab 11: Malware Analysis

Keith Schoen

CSEC3755-001

Dr. Daniel Pittman

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Contents

1	Executive Summary	1
2	Triage & File Identification	1
2.1	Creating the EICAR Test File (Sample A)	1
2.2	Basic File Identification	2
2.3	Hash Lookup on Virus Total	2
3	Static Analysis	4
3.1	String Analysis with Sysinternals Strings	4
3.2	Examine PE Structure via VirusTotal's Details Tab	5
3.3	Choose a Real Malware Sample (Sample B)	5
3.3.1	Exploring Sample B's Virus Total Page	6
4	Dynamic Analysis	9
5	IOC Report & Detection	9

1 Executive Summary

2 Triage & File Identification

Note about this section:

The Add-WindowsCapability command for Sysmon that was given did not work for this environment. Instead, I used the following:

```
# Download

Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip"
    -OutFile "$env:TEMP\Sysmon.zip"

# Extract

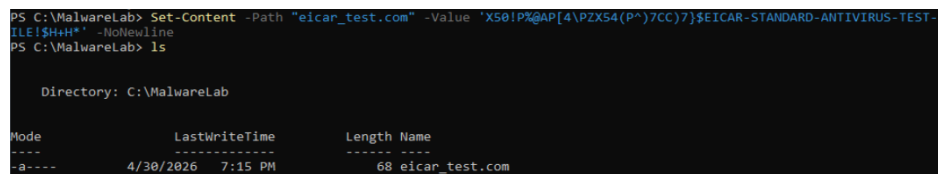
Expand-Archive -Path "$env:TEMP\Sysmon.zip" -DestinationPath "C:\Tools\
    Sysmon"

# Install (run this as Admin)

C:\Tools\Sysmon\Sysmon64.exe -accepteula -i
```

2.1 Creating the EICAR Test File (Sample A)

The following screenshots show the process of creating the EICAR test file in the Windows VM. I used the full sting given hoping to see Defender defy my expectations and catch the file at creation, but the file survived create and `Get-Content`



```
PS C:\MalwareLab> Set-Content -Path "eicar_test.com" -Value "X50!P%AP[4\pZX54(P*)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$Hh*" -NoNewline
PS C:\MalwareLab> ls

Directory: C:\MalwareLab

Mode                LastWriteTime         Length Name
----                -
-a----          4/30/2026   7:15 PM             68 eicar_test.com
```

Figure 1.1: Creation of the EICAR test file

```
PS C:\MalwareLab> Get-MpThreatDetection | Sort-Object InitialDetectionTime -Descending | Select-Object -First 3 ThreatID
, InitialDetectionTime, Resources
PS C:\MalwareLab> Get-Content .\eicar_test.com
X50!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
PS C:\MalwareLab> Get-MpThreatDetection | Sort-Object InitialDetectionTime -Descending | Select-Object -First 3 ThreatID
, InitialDetectionTime, Resources
PS C:\MalwareLab>
```

Figure 1.2: The EICAR test file survives Windows Defender

2.2 Basic File Identification

```
PS C:\MalwareLab> (Get-Item .\eicar_test.com).Length
68
PS C:\MalwareLab> Get-FileHash -Algorithm SHA256 .\eicar_test.com

Algorithm      Hash                                          Path
-----
SHA256         5B726692BEFC9CE68935B8081A1161F969DADD8DD921299416910EB3A8A1D702  C:\MalwareLab\eicar_test.com

PS C:\MalwareLab> Get-FileHash -Algorithm MD5 .\eicar_test.com

Algorithm      Hash                                          Path
-----
MD5            6A26CC095F9132549AE21EA4065899F6      C:\MalwareLab\eicar_test.com

PS C:\MalwareLab> Get-Content .\eicar_test.com
X50!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
PS C:\MalwareLab> strings.exe .\eicar_test.com

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

X50!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
PS C:\MalwareLab>
```

Figure 1.3: Results of file identification commands for the EICAR test file

2.3 Hash Lookup on Virus Total

- Out of 62 security vendors on Virus Total, 6 flagged the file as malicious.
- A few names given for the threat are:
 - Virus/EICAR_Test_File
 - NotAThreat.EICAR[TestFile]
 - Eicar_test_1
- Tags shown for the 'threat' are:
 - powershell
 - idle

- long-sleeps
- File type according to Virus Total: **Powershell**
- As for the community tab, there isn't any activity early then 10 years.

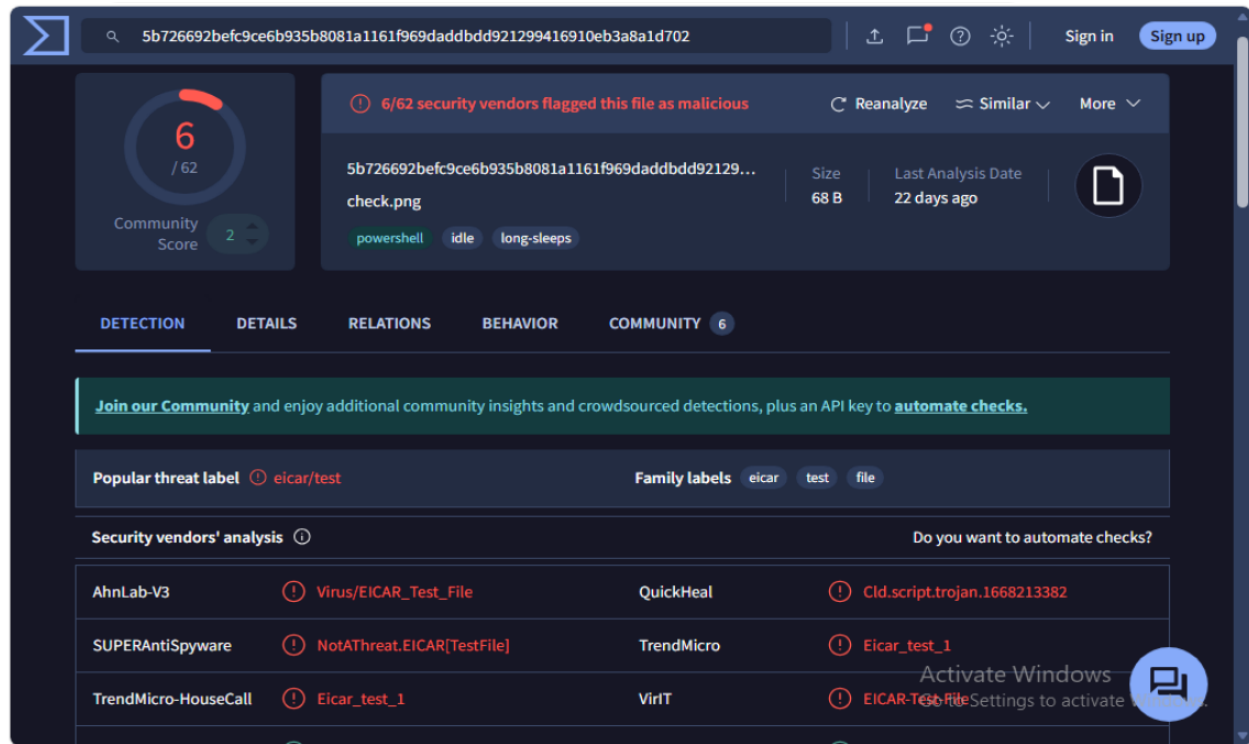


Figure 1.4: Results for Virus Total hash look-up

It should be noted that while the results on Virus Total showed a 6/62, the only results to show in the 'Security vendors' analysis' table were those from AhnLab-V3, SUPER-AntiSpyware, and TrendMicro-HouseCall.

What kind of threat is this?

This is an idle **Powershell script** that waits until the right conditions are met before executing. This would explain how it was able to evade detection from Windows Defender despite being flagged as malicious.

3 Static Analysis

3.1 String Analysis with Sysinternals Strings

Executed the following commands on the EICAR test file from section one to demonstrate how it works at a basic level:

```
# Extract all strings at least 8 characters long and save to file
strings.exe -n 8 .\eicar_test.com > eicar_strings.txt

# Open the output and examine it
notepad eicar_strings.txt
```

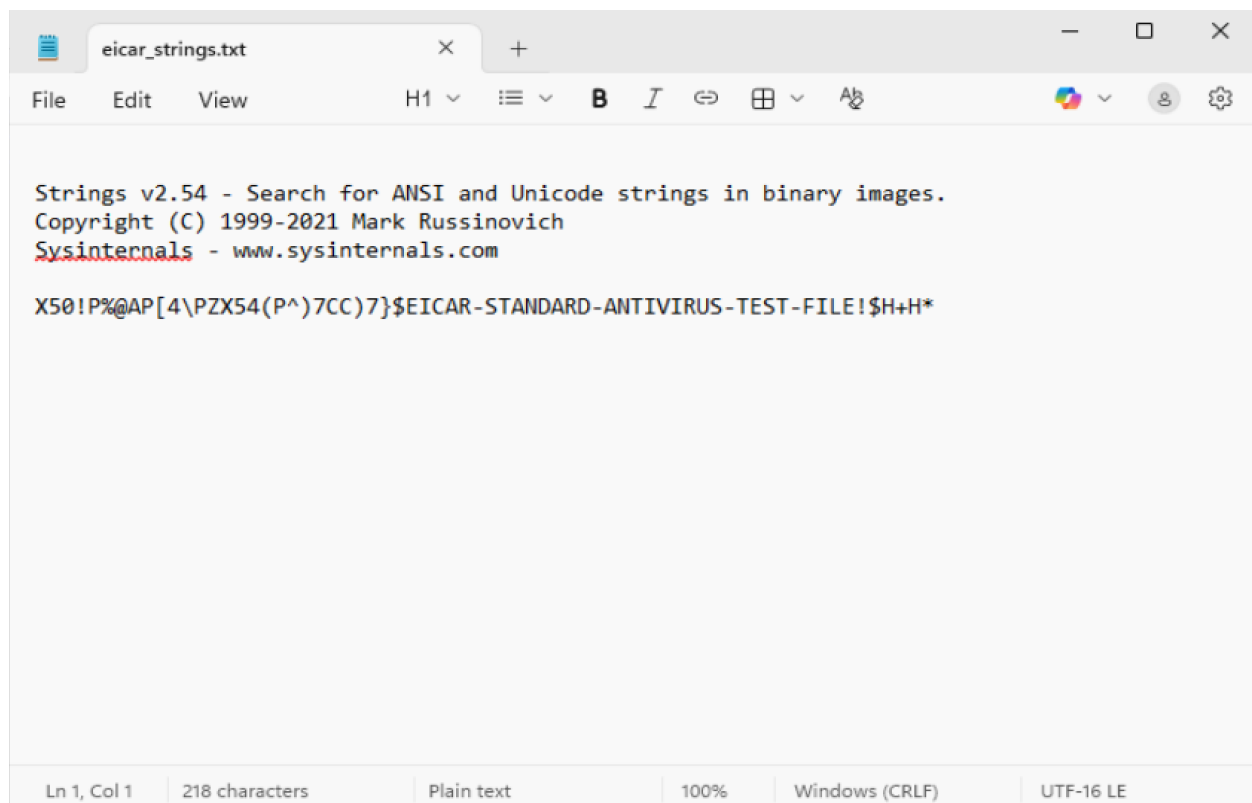


Figure 2.1: Strings output for the EICAR test file

3.2 Examine PE Structure via VirusTotal's Details Tab

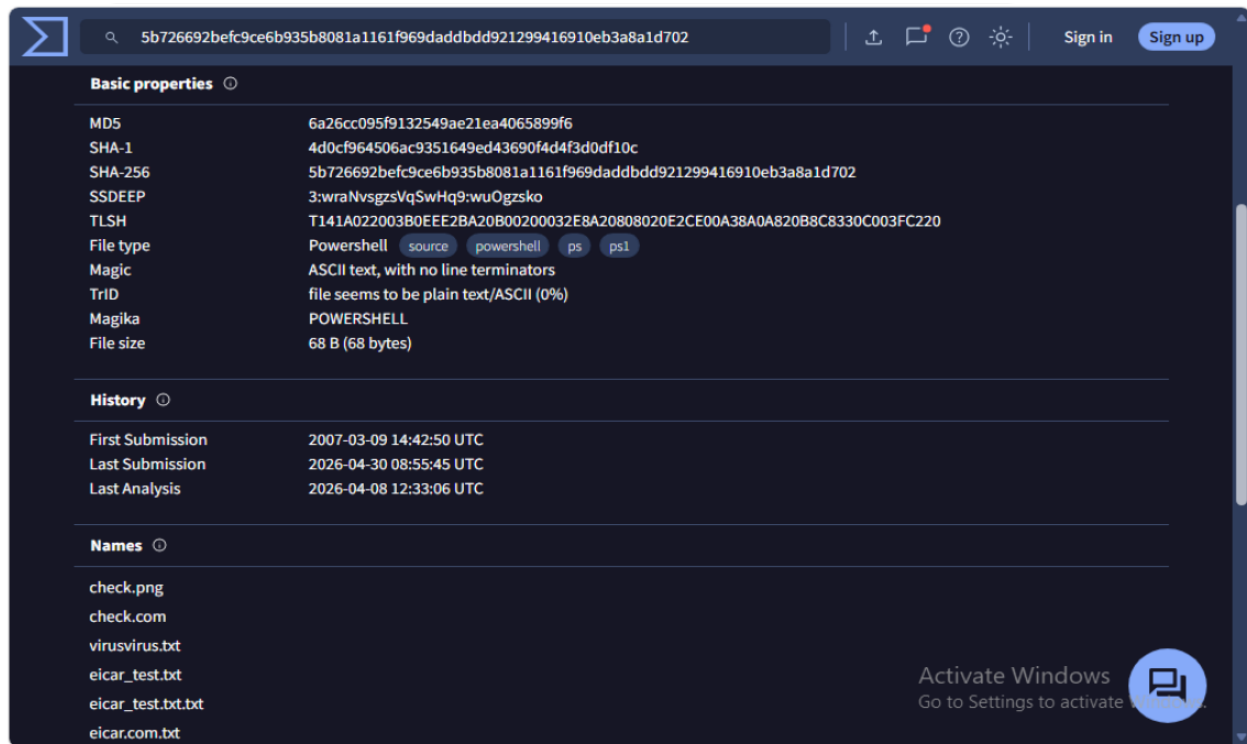


Figure 2.2: Details section for EICAR test file hash lookup

3.3 Choose a Real Malware Sample (Sample B)

Note about malware selection for Sample B

SHA256 Hash: 939b247d1cf5a8b674ff632365af9982256363d6ae390876d0d3d9cbc07d163b

Malware Family: phantomstealer

First submission to VT: 2026-04-30 07:14:26 UTC

VT Report URL:

<https://www.virustotal.com/gui/file/939b247d1cf5a8b674ff632365af9982256363d6ae390876d0d3d9cbc07d163b/summary>

Hybrid Analysis Report URL:

<https://hybrid-analysis.com/sample/939b247d1cf5a8b674ff632365af9982256363d6ae390876d0d3d9cbc07d163b/69f301586d9a87d56d06d3b7>

Honestly, I decided on a RAT because I like the animal and the malware category. When I did quick overview, this one had tons of documentation making it an easy first grab. Along with the amount of information already becoming available, it is made extra exciting in that this is brand-new submission to VT (about 3 hours old). It's a great pick for Sample B.

3.3.1 Exploring Sample B's Virus Total Page

Detection Tab

- **Detection Ratio:** 39/71
- This malware is new so there isn't a name yet, but common themes include RAT, MSIL, and Trojan. Skyhigh(SWG) seems to be the only AV that has given it a name so far: "Artemis!Trojan". The 'popular threat label' is [trojan.msil/phantomstealer](#).
- **Family:** phantomstealer

Details Tab

- **File type and size:** Win32 EXE / 876.50 KB (897536 bytes)
- **First Submission:** 2026-04-30 07:14:26 UTC
- **PE Header Information:**
 - Target Machine: Intel 386 or later processors and compatible processors
 - Compilation Timestamp: 2026-04-30 06:12:58 UTC
 - Entry Point: 902118
 - Contained Sections: 3
- **Imports:**
 - mscoree.dll

* _CorExeMain

Behavior Tab

- **Sandboxes:**

- CAPA
- VirusTotal Observer

- **Process Tree:**

840 - 939b247d1cf5a8b674ff632365af9982256363d6ae390876d0d3d9cbc07d163b.exe

276 - C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe Add-MpPreference
-ExclusionPath C:\Users\j\USER\Downloads\939b247d1cf5a8b674ff632365af9982256363d
6ae390876d0d3d9cbc07d163b.exe

2124 - ead8bccff0d208d26935d05c0ec80515.bat

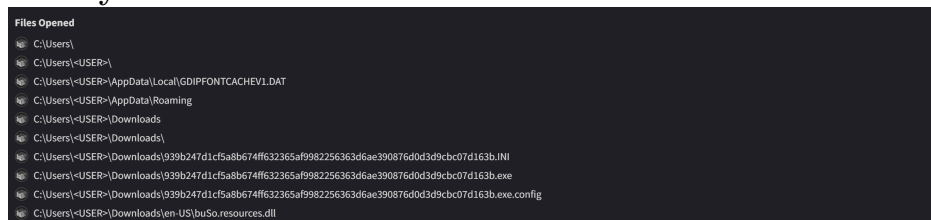
2932 - ead8bccff0d208d26935d05c0ec80515.bat

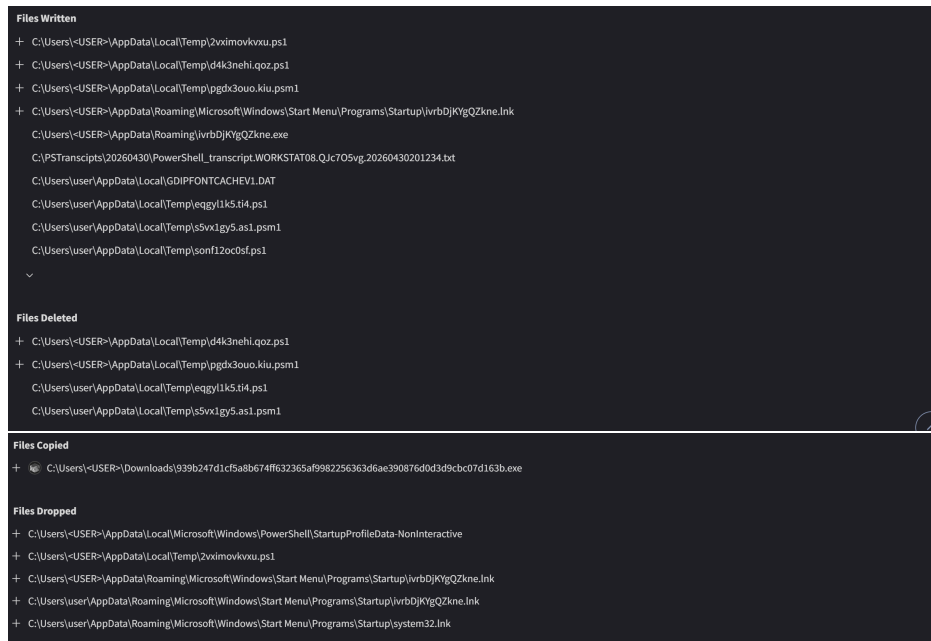
2968 - 0906e49c252cde958db247fef205b1bb.exe

892 - cde09bcd5fde1e2eac52c0f93362b79.exe

2860 - 0e058c51d287f1a00f674d8fefdf1e35.exe

- **File System Actions:**





Figures 2.3-.5: Details of Malware file operations

• Registry Actions:

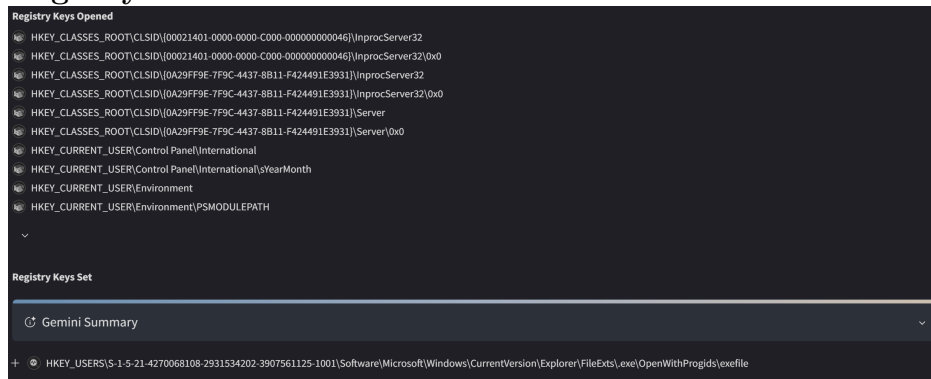


Figure 2.6: Details of Malware registry actions

• Network Communications

– IP Traffic

* UDP 8.8.8.8:53

* TCP 199.232.210.172:80

• MITRE ATT&CK Techniques:

Execution	Persistence	Privilege Escalation	Stealth	Credential Access
TA0002 4 Techniques	TA0003 5 Techniques	TA0004 3 Techniques	TA0005 6 Techniques	TA0006 2 Techniques
Scheduled Task/Job T1053 2	New Service T1050 1	New Service T1050 1	Obfuscated Files or Infor... T1027 1	Input Capture T1056
Command and Scripting Inte... T1059 1	Scheduled Task/Job T1053 2	Scheduled Task/Job T1053 2	Deobfuscate/Decode Files or ... T1140 1	Hooking T1179
PowerShell T1086 2	Registry Run Keys / Startup F... T1060 1	Hooking T1179 1	Hide Artifacts T1564	
Shared Modules T1129 3	Modify Registry T1122 1		Reflective Code Loading T1620 1	
	Hooking T1179 1			

Figure 2.7: Table of MITRE ATT&CK techniques used

Relations Tab

- **Contacted IPs:**

IP	Detections	Autonomous System	Country
199.232.210.172	0 / 91	54113	US
8.8.8.8	0 / 91	15169	US

- **Dropped Files:**

Scanned	Detections	File type	Name
2026-04-30	0 / 62	Windows shortcut	system32.lnk
2026-04-30	21 / 62	Windows shortcut	lvrbDjKqQ2kne.lnk
2026-04-30	39 / 71	Win32 EXE	TT-PAYMENT COPY -04.bat
2026-04-30	0 / 61	?	StartupProfileData-NonInteractive
?	?	file	03b3383a80e7e9d3b2543d19fcd557de85664e7500efd2cf86120f4d8b0
?	?	file	0d5b7d1dc823364ff41d6fcafe9f76b9e1290ebbf387a3b6a2ff4c655a76c
?	?	file	0835269f16770370277ad215a8c50cf4924a3265a7e08c07ebba2fbeb31e
?	?	file	445c2baeb30c50743737052b1880c3c413be0b22fce790c9dd449835dbe0400b
?	?	file	56acc3b428c2561797056a2c367ed5e9501aba2262bb4030cb6706c719abda5f
?	?	file	7267c26e6eca#dd0118a0eb8cc2f62fb4195f7d8c716cb7873d812f4d6f741a

- **Related Samples:**

Execution Parents (2)			
Scanned	Detections	Type	Name
2026-04-30	19 / 62	Email	Templ[217].eml
2026-04-30	39 / 71	Win32 EXE	TT-PAYMENT COPY -04.bat
Bundled Files (4)			
Scanned	Detections	File type	Name
2026-04-28	0 / 61	XML	.rsrc/MANIFEST/1
?	?	file	.reloc
?	?	file	.rsrc/version.txt
?	?	file	.text

4 Dynamic Analysis

5 IOC Report & Detection

++report written and formatted with help from Claude (Sonnet 4.6)++